
Algorithm 1: Bilateral ZUPT stride-length estimation.

Input : Ipsilateral 3-axis accelerometer (*accel*) and gyroscope (*gyro*)
Contralateral toe-off times (*contra_to*)
Sampling rate (*fs*)

Output : Stride lengths (*stride_lengths*)

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/* Step 1. Stance detection with contralateral validation */
1 gyro_norm = magnitude of gyro at each sample
2 accel_var = 40 ms sliding variance of accel
3 gyro_thresh = clip(15th percentile of gyro_norm,
4 0.3, 2.0 rad/s)
5 accel_thresh = clip(25th percentile of accel_var,
6 0.2, 5.0 (m/s2)2)
7 candidates = samples where gyro_norm < gyro_thresh and accel_var <
  accel_thresh
8 for each candidate segment do
9   if any contra_to event falls within  $\pm 150$  ms of the segment boundary then
10    | accept segment as stance
11   else
12    | reject segment
13   end
14 end
15 recover missed stances near unmatched contra_to using relaxed gyroscope-only threshold
  (25th percentile, 0.5 rad/s floor)
16 reject stance segments < 30 ms, merge segments separated by < 75 ms, split strides > 1.8 s at a
  mid-stride quiet period
/* Step 2. Per-stride integration with retroactive
  gravity-residual correction */
17 estimate gyro_bias and accel_bias from stance samples of each walking bout
18 for each stride between consecutive stance midpoints do
19   track orientation with Mahony AHRS (gain 1.0 during stance, 0.05 during swing)
20   rotate (accel – accel_bias) into the navigation frame, subtract gravity
21   measure gravity residual at the next stance onset
22   subtract a linear ramp of that residual through the preceding swing (gain 0.5)
23   integrate corrected acceleration  $\rightarrow$  velocity, distribute the accumulated velocity error
    linearly back across the swing, then set velocity=0 at stance ends
24   integrate velocity  $\rightarrow$  position
25   stride_length = horizontal distance between stride start and stride end positions
26   reject stride unless duration 0.4–2.0 s, length 0.3–2.5 m, peak gyro_norm > 3.0 rad/s
27 end
28 return stride_lengths
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